11/3/25

MODULE-2

Permutations and Combinations

→ An ordered arrangement of r elements of a set containing n distinct element is called a permutation.

Denoted by $P(n, r)$ or ${}^{n}P_{r}$ when $r \leq n$.

→ An unordered selection of r elements of a set containing n distinct elements is called a combination.

Denoted by $C(n, r)$ or ${}^{n}C_{r}$ or $\binom{n}{r}$.

→ ${}^{n}P_{r} = \frac{n!}{(n-r)!}$ ${}^{n}P_{n} = n! = {}^{n}P_{n-1}$ ${}^{n}P_{1} = n$ ${}^{n}P_{0} = 1$

→ ${}^{n}C_{r} = \frac{n!}{(n-r)!\,r!}$ ${}^{n}C_{n} = 1$ ${}^{n}C_{1} = n$ ${}^{n}C_{0} = 1$

→ Product rule :-

$$ {}^{n}P_{r} = {}^{n}C_{r} \cdot r! $$

→ Pascal's identity :-

$$ {}^{n}C_{r-1} + {}^{n}C_{r} = {}^{n+1}C_{r} $$

→ $\frac{{}^{n}C_{r}}{{}^{n-1}C_{r-1}} = n/r$ $\frac{{}^{10}C_{5}}{{}^{9}C_{4}} = 10/5 = \underline{\underline{2}}$

→ $\frac{{}^{n}C_{r}}{{}^{n}C_{r-1}} = \frac{n-r+1}{r} = \frac{{}^{10}C_{5}}{{}^{10}C_{4}} = \frac{10-5+1}{5} = \underline{\underline{6/5}}$

13/3/25 . **Permutations with repetition**

→ The number of permutation of n different objects taken r at a time, when repetition is allowed is $n \times n \times n \times \cdots$ r times $= n^r$.

→ The number of permutations of n things taken all at a time when p of them are similar of one type; q of them are similar of second type, r of them are similar of third type and the remaining so on ... is $\frac{n!}{p!\,q!\,r!}$

Eg: AAA BBBB CC $\frac{9!}{3! \times 4! \times 2!}$

→ n distinct items in which r particular items should come in an order not necessary together is $\frac{n!}{r!}$

Example 6.1.

a). Assuming that repetitions are not permitted, how many four-digit numbers can be formed from the six digits 1, 2, 3, 5, 7, 8?

ans) $^6C_4 = \frac{6 \times 5 \times 4 \times 3}{1 \times 2 \times 3 \times 4} = \cancel{15}$

6 ways 5 ways 4 ways 3 ways

$6 \times 5 \times 4 \times 3 = \underline{\underline{360}}$

b) How many of these numbers are less than 4000?

ans:)

3 6 4 3

(1, 2, 3)

$3 \times 5 \times 4 \times 3 = \underline{\underline{180}}$

c) How many of the numbers in part (a) are even?

ans.) $\underset{3}{\_}\ \underset{4}{\_}\ \underset{5}{\_}\ \underset{\substack{2\\(2,8)}}{\_}$ $\quad 3\times4\times5\times2 = \underline{\underline{120}}$

d) How many of the numbers in part (a) are odd?

ans.) $\underset{3}{\_}\ \underset{4}{\_}\ \underset{5}{\_}\ \underset{\substack{4\\(1,3,5,7)}}{\_}$ $\quad 3\times4\times5\times4 = \underline{\underline{240}}$

e) How many of the numbers in part (a) are multiple of 5?

ans.) $\underset{3}{\_}\ \underset{4}{\_}\ \underset{5}{\_}\ \underset{(5)}{\_}\ 1$ $\quad 3\times4\times5\times1 = \underline{\underline{60}}$

f) How many of the numbers in part (a) contain both the digits 3 and 5?

ans) $\underline{3}\ \underline{5}\ \underset{4}{\_}\ \underset{3}{\_}$ $\quad 4\times3 = 12$

$\underline{5}\ \underline{3}\ \underset{4}{\_}\ \underset{3}{\_}$ $\quad 4\times3 = 12$ .... $4P_2$ arrangements of 5, 3 is possible $= \underline{\underline{12}}$.

$12\times12 = \underline{\underline{144}}$

Example 6.2:

a) In how many ways can 6 boys and 4 girls sit in a row?

ans.) 10! ways (6+4 = 10) 10! arrangements.

b) In how many ways can they sit in a row if the boys are to sit together and the girls are to sit together?

ans.) No. of ways the ~~girls~~ boys can be arranged = 6!

No. of ways the ~~boys~~ girls can be arranged = 4!

By assuming the boys as one unit and the girls as another, these 2 units can be arranged in 2! = 2 ways.

$\therefore 2! \times 6! \times 4! = \underline{\underline{34560}}$

6!	4!
4!	6!

2 ways.

c) In how many ways can they sit in a row if the girls are to sit together?

ans.) [- - - - -] (1) fixed. _ _ _ _ _ _ (1 2 3 4 5 6; 6+1 = 7)

[- - - - -] → 4! ($^4P_4$)

$= 7! \times 4! = \underline{\underline{1,20,960}}$

d) In how many ways can they sit in a row if just the girls are to sit together?

ans.) No. of ways in which girls only sit together = (No. of ways in which girls sit together) − (No. of ways in which boys sit together and girls sit together)

$= 120960 - 34560 = \underline{\underline{86400}}$

Example 6.3

How many different paths in the xy plane are there from (1,3) to (5,6) if a path proceeds one step at a time by going either one step to the right (R) or one step upward (U)?

ans.) To reach the point (5,6) from (1,3), one has to traverse 5−1 = 4 steps to the right and 6−3 steps to the up. Hence the total number of 7 steps consist of 4 R's and 3 U's.

To traverse the paths, one can take R's and U's in any order. Hence, the required number of different path is equal to the number of permutations of 7 steps of which 4 are of the same type (namely R) and 3 are of the same type (namely U).

Required number of paths $= \frac{7!}{4! \times 3!} = \underline{\underline{35}}$.

14/3/25 ·) Sum rule

If r activities can be performed in $n_1, n_2, \dots, n_r$ ways and if they are disjoint. Viz., cannot be performed simultaneously, then any one of the r activities can be performed in $(n_1 + n_2 + \dots n_r)$ ways.

Eg: How many positive integers n can be formed using the digits 3, 4, 4, 5, 5, 6, 7, if n has to exceed 50,00,000?

ans:) $\underset{\text{fixed}}{5}$ _ _ _ _ _ _
6 5 4 3 2 1

$\frac{6!}{2!} = \underline{\underline{360}}$.
↳ (4,4) Repetition.

$\underset{\text{fixed}}{6}$ _ _ _ _ _ _ $\frac{6!}{\underbrace{2!}_{(4,4)} \times \underbrace{2!}_{(5,5)}} = \underline{\underline{180}}$.

$\underset{\text{fixed}}{7}$ _ _ _ _ _ _ $\frac{6!}{2! \times 2!} = \underline{\underline{180}}$

∴ No. of numbers exceeding 50,00,000 $= 360 + 180 + 180 = \underline{\underline{720}}$.

Q) How many bit strings of length 10 contain (a) exactly four 1's (b) atmost four 1's (c) at least four 1's (d) an equal number of 0's and 1's?

ans) (a) String length = 10

exactly four 1's $\Rightarrow$ six 0's

No. of required bit strings = $\frac{10!}{4! \times 6!} = \underline{\underline{210}}$

(b) atmost four 1's $\Rightarrow \frac{10!}{0! \times 10!} + \frac{10!}{1! \times 9!} + \frac{10!}{2! \times 8!} + \frac{10!}{3! \times 7!} + \frac{10!}{4! \times 6!}$

$= \underline{\underline{386}}$

(c) At least four 1's $\Rightarrow \frac{10!}{4! \times 6!} + \frac{10!}{5! \times 5!} + \frac{10!}{6! \times 4!} + \frac{10!}{7! \times 3!} + \frac{10!}{8! \times 2!} \times \frac{10!}{9! \times 1!} \times \frac{10!}{10! \times 0!}$

$= \underline{\underline{848}}$

(d) Equal number of 0's & 1's $\Rightarrow \frac{10!}{5! \times 5!} = \underline{\underline{252}}$.

Q) How many permutations of the letters ABCDEFG contain, (a) the string BCD

(b) the string CFGA

(c) the strings BA and GF

(d) the strings ABC and DE

(e) the strings ABC and CDE

(f) the strings CBA and BED

ans) (a) $\boxed{BCD}$ _ _ _ _ $= 5! = 120$

(b) $\boxed{CFGA}$ $\underline{2}\ \underline{3}\ \underline{4}$ $\quad 4! = 24$

(c) $\overset{1}{\boxed{BA}}\ \overset{2}{\boxed{GF}}$ $\underline{3}\ \underline{4}\ \underline{5}$ $\quad 5! = 120$

(d) $\overset{1}{\boxed{ABC}}\ \overset{2}{\boxed{DE}}$ $\underline{3}\ \underline{4}$ $\quad 4! = 24$

(e) $\boxed{ABC}\ \boxed{CDE}$

ordered

$\overset{1}{\boxed{ABCDE}}$ $\underline{2}\cdot\underline{3}$ $\quad 3! = 6$

Even though (ABC) and (CDE) are two strings, they contain the common letter C. If we include the strings (ABCDE) in the permutation it includes both the strings (ABC) and (CDE). Moreover we cannot use the letter C twice. Hence we have to permute the 3 objects (ABCDE), F and G. This can be done in $3! = 6$ ways.

(f) $\boxed{CBA}\ \boxed{BED}$

unordered

~~ABCDE~~

$\boxed{CBABED}$

To include the 2 strings (CBA) and (BED) in the permutations, we require the letter B twice, which is not allowed.

Hence, the require no. of permutation = 0.

a) If 6 people A, B, C, D, E, F are seated about a round table. how many different circular arrangements are possible, if arrangements are considered the same when one can be obtained from the other by rotation? If A, B, C are females and the others are males, in how many arrangements do the sexes alternate?

ans.) the no. of circular permutations of n different things taken all at a time is $(n-1)! = \frac{{}^nP_n}{n} = \frac{n!}{n}$

∴ The required no. of circular arrangements $= (6-1)!$
$= 5! = \underline{\underline{120}}.$

Note :-
In case of flipping (garlands, ring ...)

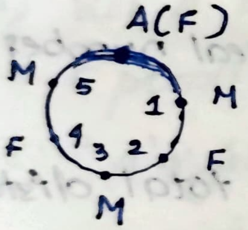
→ No. of circular permutation taken all at a time $= \dfrac{(n-1)!}{2}$

$= \dfrac{{}^nP_n}{2n} = \dfrac{n!}{2n}$

→ No. of circular permutations taken r at a time

$= \dfrac{{}^nP_r}{r}$

→ In case of flipping $\dfrac{{}^nP_r}{2r}$



since rotation does not alter the circular arrangement, we can assume that A occupies the top position as shown in the figure ∴

Of the remaining places, position 1, 3, 5 must be occupied by the 3 males. This can be achieved in $P(3,3) = 3! = 6$ ways.

The remaining two places 2 and 4 should be occupied by the remaining two females. This can be achieved in $P(2,2) = 2$ ways.

∴ Total no. of required circular managements $= 6 \times 2$
$= \underline{\underline{12}}.$

18/3/25 •) Pigeonhole principle

→ If n pigeons are accommodated in m pigeon-holes and $n>m$ then at least one pigeonhole will contain two or more pigeons. Equivalently, if n objects are put in m boxes and $n>m$, then at least one box will contain two or more objects.

→ If n pigeons are accommodated in m pigeonholes and $n>m$, then one of the pigeonholes must contain atleast $\left\lfloor \frac{(n-1)}{m} \right\rfloor + 1$ pigeons, where $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x, which is a real number.

Eg:) A man hiked for 10 hours and covered a total distance of 45 Km. It is known that he hiked 6 Km in the first hour and only 3 Km in the last hour. Show that he must have hiked at least 9 km within a certain period of 2 consecutive hours.

ans) Since, the man hiked $6+3=9$ km in the first and last hours, he must have hiked $45-9=36$ km during the period from second to ninth hours.

If we combine the second and third hours together, the fourth and fifth hours together etc.. and the eighth and ninth hours together, we have 4 time periods. Let us now treat 4 time periods as pigeonholes and 36km as 36 pigeons. Using the generalised pigeonhole principal, the least no of pigeons accommodated in one pigeonhole = $\left\lfloor \frac{36-1}{4} \right\rfloor + 1 = \lfloor 8.75 \rfloor + 1 = 9$

viz., the man must have hiked at least 9km in one time period of 2 consecutive hours.

Q) Prove that in any group of six people, at least three must be mutual friends or at least three must be mutual strangers.

Ans) Let A be one of the six people. Let the remaining 5 people be accommodated in 2 rooms labeled "A's friends and strangers to A. Treating 5 people as 5 pigeons and 2 rooms as pigeonholes, by the generalized pigeonhole principles one of the rooms must contain $\left\lfloor \frac{5-1}{2} \right\rfloor + 1 = 3$ people.

Let the room labeled A's friends contain 3 people. If any two of these 3 people are friends, then together with A, we have a set of 3 mutual friends. If no two of these 3 people are friends, then these 3 people are mutual strangers. In either case, we get the required conclusion.

If the room labeled "strangers to A" contain 3 people. we get the required conclusion by similar argument.

Q) A bag contains 10 red marbles, 10 white marbles, and 10 blue marbles. What is the minimum no. of marbles you have to choose randomly from the bag to ensure that we get 4 marbles of same colour?

ans.) $\left\lfloor \frac{x-1}{3} \right\rfloor + 1 \geq 4 \rightarrow$ needed balls

no. of colours

$\frac{x-1}{3} \geq 3$

$\cancel{\frac{x}{3}} \not\geq \cancel{4}$ $\quad x-1 \geq 9$

$\cancel{x}$ $\quad x \geq 10$

Q) If a Martian has an infinite number of red, blue, yellow and black socks in a drawer. What is the minimum numbers of socks that the Martian must pull out of the drawer to guarantee they have a pair?

ans) Draw 1:) Red (consider)

Draw 2:) Worst case Blue

Draw 3:) Worst case Yellow

Draw 4:) Worst case Black

Now we have one sock of each pair so in the 5^{th} draw it will match any one of the colour and will get a pair.

$\therefore$ The minimum no. of socks to be pulled out $= \underline{\underline{5}}$

19/3/25 . Principle of inclusion - exclusion

If A and B are finite subsets of a finite universal set U, then $|A \cup B| = |A| + |B| - |A \cap B|$, where $|A|$ denotes the cardinality of (the number of elements in) the set A.

This principle can be extended to a finite number of finite sets $A_1, A_2, \dots A_n$ as follows:

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \dots$$

$$\dots (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|,$$

where the first sum is over all i, the second sum is over all pairs i, j with $i<j$; the third sum is over all triples i, j, k with $i<j<k$ and so on.

$\rightarrow n(A \cap B \cap C) = n(A) + n(B) + n(C) - n(A \cup B) - n(A \cup C) - n(B \cup C)$
$+ n(A \cup B \cup C)$

Q) There are 250 students in an engineering college. Of these 188 have taken a course in fortran, 100 have taken a course in C and 35 have taken a course in Java. Further 88 have taken courses in both fortan and C. 23 have taken courses in both C and Java and 29 have taken courses in both fortan and Java. If 19 of these students have taken all the three courses, how many of these 250 students have not taken a course in any of these three programming languages?

ans:) Let F, C, and J denote the students who have taken the languages fortan, C and Java respectively.

Then $|F| = 188$ $\quad |F \cap C \cap J| = 19$.

$|C| = 100$

$|J| = 35$

$|F \cap C| = 88$,

$|C \cap J| = 23$

$|F \cap J| = 29$

Then the no. of students who have taken at least one of the three languages is given by

$$|F \cup C \cup J| = |F| + |C| + |J| - [F \cap C| - |C \cap J| - |F \cap J| + |F \cap C \cap J|$$

$$= (188 + 100 + 35) - (88 + 23 + 29) + 19$$

$$= \underline{\underline{202}}.$$

No. of students who have not taken a course in any of these languages $= 250 - 202 = \underline{\underline{48}}$.

Q) Find the number of integers between 1 and 250 both inclusive that are not divisible by any of the integers 2, 3, 5 and 7? [Let A, B, C, D be the sets of integers that lie between 1 and 250 and that are divisible by 2, 3, 5 and 7 respectively] ans:) $|A| = \left\lfloor \frac{250}{2} \right\rfloor = 125$

Note :-

1-10

no. of elements divisible by 3

$= \left\lfloor \frac{10}{3} \right\rfloor = 3$ (3, 6, 9)

→ floor division

$|B| = \left\lfloor \frac{250}{3} \right\rfloor = 83$

$|C| = \left\lfloor \frac{250}{5} \right\rfloor = 50$

$|D| = \left\lfloor \frac{250}{7} \right\rfloor = 35$

The set of integers between 1 and 250 which are divisible by 2 and 3, viz., $A \cap B$ is the same as that which is divisible by 6, since 2 and 3 are respectively.

$\therefore |A \cap B| = \left\lfloor \frac{250}{\underbrace{6}_{2\times 3}} \right\rfloor = 41$

Similarly, $|A \cap C| = \left\lfloor \frac{250}{\underbrace{10}_{2\times 5}} \right\rfloor = 25$ $\quad |C \cap D| = \left\lfloor \frac{250}{35} \right\rfloor = 7$

$|A \cap D| = \left\lfloor \frac{250}{14} \right\rfloor = 17$ $\quad |A \cap B \cap C| = \left\lfloor \frac{250}{30} \right\rfloor = 8$

$|B \cap C| = \left\lfloor \frac{250}{15} \right\rfloor = 16$ $\quad |A \cap B \cap D| = \left\lfloor \frac{250}{42} \right\rfloor = 5$

$|B \cap D| = \left\lfloor \frac{250}{21} \right\rfloor = 11$ $\quad |A \cap C \cap D| = \left\lfloor \frac{250}{70} \right\rfloor = 3$

$|B \cap C \cap D| = \left\lfloor \frac{250}{105} \right\rfloor = 2$ $\quad |A \cap B \cap C \cap D| = \left\lfloor \frac{250}{210} \right\rfloor = 1$

$\therefore$ The number of integers between 1 and 250 that are divisible by atleast one of 2, 3, 5 and 7 is given by

$$|A \cup B \cup C \cup D| = \{|A| + |B| + |C| + |D|\} - \{|A \cap B| + \cdots + |C \cap D|\} + \{|A \cap B \cap C| + \cdots + |B \cap C \cap D|\} - \{|A \cap B \cap C \cap D|\}$$

$$= (125 + 83 + 50 + 35) - (41 + 25 + 17 + 16 + 11 + 7) + (8 + 5 + 3 + 2) - 1$$

$$= 293 - 117 + 18 - 1 = \underline{\underline{193}}$$

$\therefore$ Number of integers between 1 and 250 that are not divisible by any of the integers 2, 3, 5 and 7

= Total no. of integers − $|A \cup B \cup C \cup D|$

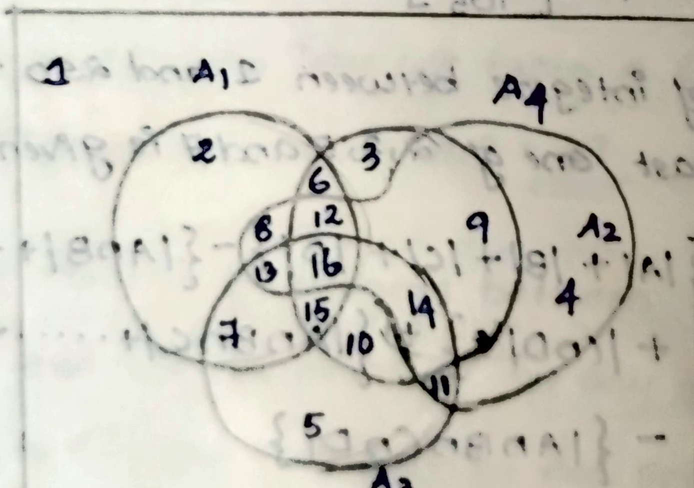
= $250 - 193 = \underline{\underline{57}}$

Q) A_1, A_2, A_3 and A_4 are subsets of a set U containing 75 elements with the following properties. Each subset contains 28 elements; the intersection of any two of the subsets contains 12 elements; the intersection of any three of the subsets contain 5 elements; the intersection of all four subsets contains 1 element.

(a) How many elements belong to none of the four subsets?

(b) How many elements belong to exactly one of the four subsets?

(c) How many elements belong to exactly two of the four subsets?



ans) a) No. of elements that belong to at least one of the four subsets = $|A_1 \cup A_2 \cup A_3 \cup A_4|$

$= \Big[\{|A_1| + |A_2| + |A_3| + |A_4|\} - \{|A_1 \cap A_2| + |A_1 \cap A_3| + |A_1 \cap A_4|$

$+ |A_2 \cap A_3| + |A_2 \cap A_4| + |A_3 \cap A_4|\} + \{|A_1 \cap A_2 \cap A_3| +$

$|A_1 \cap A_2 \cap A_4| + |A_1 \cap A_3 \cap A_4| + |A_2 \cap A_3 \cap A_4|\}$

$- |A_1 \cap A_2 \cap A_3 \cap A_4|\Big]$

$= [4 \times 28 - 6 \times 12 + 4 \times 5 - 1] = 59$

$\therefore$ No. of elements that belong to none of the four subset

$= 75 - 59 = \underline{\underline{16}}$.

b) With reference to the Venn diagram given

$= n[(2)]$

$= n(A_1) - [n(6) + n(7) + n(8) + n(12) + n(13) + n(15) + n(16)]$

$= n(A_1) - \Big[\{n(6) + n(12) + n(15) + n(16)\} + \{n(7) + n(13) + n(15)$

$+ n(16)\} + \{n(8) + n(12) + n(13) + n(16)\} - n(12) - n(13) -$

$n(15) - 2n(16)\Big]$

$= n(A_1) - [n(A_1 \cap A_2) + n(A_1 \cap A_3) + n(A_1 \cap A_4)] +$

$[n(A_1 \cap A_2 \cap A_4) + n(A_1 \cap A_3 \cap A_4) + n(A_1 \cap A_2 \cap A_3)]$

$- 2n[(A_1 \cap A_2 \cap A_3 \cap A_4)]$

$= 28 - 3 \times 12 + 3 \times 5 + 2 \times 1$

$= \underline{\underline{9}}$

Similarly $n(A_2 \text{ alone}) = n(A_3 \text{ alone}) = n(A_4 \text{ alone}) = 9$

$\therefore$ No. of elements that belong to exactly one of the subsets $= 36$.

c) With reference to the Venn diagram

$n(A_1 \text{ and } A_2) \text{ only} = n(6)$

$= n(A_1 \cap A_2) - \{n(15) + n(16)\} - \{n(12) + n(16)\} + n(16)$

$= n(A_1 \cap A_2) - n(A_1 \cap A_2 \cap A_3) - n(A_1 \cap A_2 \cap A_4)$

$+ n(A_1 \cap A_2 \cap A_3 \cap A_4)$

$= 12 - 5 - 5 + 1 = \underline{\underline{3}}$

Similarly $n(A_1 \text{ and } A_3 \text{ only}) = n(A_1 \text{ and } A_4 \text{ only})$

$= n(A_2 \text{ and } A_3 \text{ only}) = n(A_2 \text{ and } A_4 \text{ only}) = n(A_3 \text{ and } A_4 \text{ only}) = 3$

$\therefore$ No. of elements that belong to exactly two of the subsets $= 18$

Tutorial

5A) How many solutions does the equation $x_1 + x_2 + x_3 = 13$ have, where x_1, x_2, x_3 are non-negative integers less than 6? Use the principle of inclusion-exclusion.

Ans) $x_1 + x_2 + x_3 + \cdots \cdots x_m = k$

No. of possible solutions

$= C(m+k-1, k)$

$x_1+x_2+x_3=13$ $\quad (0 \le x_1, x_2, x_3 \le 5)$

$x_1 < 6,\ x_2 < 6,\ x_3 < 6 \Rightarrow$ $\cancel{x_1 \le 5, x_2 \le 5, x_3 \le 5}$

Total no. of solutions $= C(n+k-1, k)$

$n = 3 \quad (x_1+x_2+x_3)$

$k = 13$

$\Rightarrow C(3+13-1, 13) = {}^{15}C_{13} = \underline{\underline{105}}$

$N(P_1)$ $\quad x_1 = 6$

$x_1 \ge 6$ $\quad x_2+x_3 = 13-6 = 7$ $\quad x_1'+x_2+x_3 = 7$

no. of solution with $x_1 \ge 6$ then the no. of solutions $= C(3+7-1, 7) = {}^{9}C_{7} = \underline{\underline{36}}$

By symmetry, the same no. of solutions exist if $x_2 \ge 6$ or $x \ge 6$, so the total no. of solutions $= 3 \times 36 = \underline{\underline{108}}$

$N(P_1, P_2) =$ $x_1 \ge 6$, $x_2 \ge 6$ $\quad x_1+x_2 = 12$, $6+6 = 12$ $\quad x_1'+x_2'+x_3 = 1$

No. of solutions with $x_1 \ge 6$ and $x_2 \ge 6$

$\cancel{x_1+x_2}+x_3 = 13-12 = \underline{\underline{1}}$

then the no. of solutions $= C(3+1-1, 1) = 3C_1 = \underline{\underline{3}}$

The same no. of solutions exist for the pairs $N(P_1, P_3)$ and $N(P_2, P_3)$ so the total number of solutions $= 3 \times 3 = \underline{\underline{9}}$

$N(P_1, P_2, P_3) =$ $x_1 \ge 6$, $x_2 \ge 6$, $x_3 \ge 6$ $\quad 6+6+6 = 18$ $\quad x_1+x_2+x_3 = 18$

No. of solutions with $x_1 \ge 6, x_2 \ge 6, x_3 \ge 6$

$x_1+x_2+x_3 = 13$

$x_1'+x_2'+x_3' = 13-18 = -5$

Since there are no non-negative integer $\Rightarrow$ no solution

$\therefore$ Using inclusion and exclusion, the total no. of solution is $105 - 108 + 9 = \underline{\underline{6}}$.

$N(P_1', P_2', P_3') = N - N(P_1) - N(P_2) - N(P_3) + N(P_1P_2) + N(P_1P_3) + N(P_2P_3) - N(P_1P_2P_3)$

55) A total of 1232 Students have taken a course in Tamil, 879 have taken a course in English and 114 have taken a course in Hindi. Further, 103 have taken courses in both Tamil and English, 23 have taken courses in both Tamil and Hindi and 14 have taken courses in both English and Hindi. If 2092 students have taken at least one of Tamil, English and Hindi, How many students have taken a course in all the three languages?

ans.) $|T| = 1232$ $\quad |E| = 879$ $\quad |H| = 114$ $\quad |T \cap E| = 103$

$|T \cap H| = 23$ $\quad |E \cap H| = 14$ $\quad |T \cup E \cup H| = 2092$

$|T \cap E \cap H| = ?$

$\cancel{|T \cap E \cap H| = |T| + |E| + |H| - |T \cap E| - |T \cap H| - |E \cap H|}$
$\cancel{+ |T \cup E \cup H|}$

$= \cancel{1232 + 879 + 114 - 103 - 23 - 14 + 2092}$

$=$

$$|T \cup E \cup H| = |T| + |E| + |H| - |T \cap E| - |T \cap H| - |E \cap H| + |T \cap E \cap H|$$

$$2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |T \cap E \cap H|$$

$$2092 - 2085 = |T \cap E \cap H|$$

$$|T \cap E \cap H| = 7 //$$

20/3/25 •) Dearrangement

the number of dearrangements of a set of n elements is given by
$$D_n = n!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\cdots+(-1)^n\frac{1}{n!}\right].$$

A dearrangement is a permutation of objects in which no object occupies its original position. For example the dearrangements of 123 are 231 and 312. viz., $D_3 = 2$.

21453 is a dearangement of 12345; but 21543 is not a dearrangement of 12345, since 4 occupies its original position.

Eg:) Five gentlemen A, B, C, D and E attend a party, where before joining the party, they leave their overcoats in a cloak room. After the party, the overcoats get mixed up and are returned to the gentleman in a random or manner. Using the principle of inclusion-exclusion, find the probability that none receives his own overcoat.

ans:) Required probability = $\dfrac{\text{No. of permutations in which none gets his overcoat}}{\text{No. of all possible permutations of the coats}}$

$$= \frac{D_5}{5!} = \frac{5!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}\right]}{5!}$$

$$= 11/30.$$

21/3/25 •) **Binomial theorem**

→ Pascal's triangle

$$
\begin{array}{ccccccccccc}
 & & & & & 1 & & & & & \\
 & & & & 1 & & 1 & & & & \\
 & & & 1 & & 2 & & 1 & & & \\
 & & 1 & & 3 & & 3 & & 1 & & \\
 & 1 & & 4 & & 6 & & 4 & & 1 & \\
1 & & 5 & & 10 & & 10 & & 5 & & 1
\end{array}
$$

→ Binomial theorem for any positive integer n

$n-1+1=n$

$$(a+b)^n = {}^nC_0\, a^n + {}^nC_1\, a^{n-1} b^1 + {}^nC_2\, a^{n-2} b^2 + \cdots + {}^nC_{n-1}\, a b^{n-1} + {}^nC_n\, b^n$$

Eg:) Expand $(2x+3y)^4$ by binomial theorem

ans:) $(2x+3y)^4 = {}^4C_0(2x)^4 + {}^4C_1(2x)^3(3y) + {}^4C_2(2x)^2(3y)^2 + {}^4C_3(2x)(3y)^3$
$+ {}^4C_4(3y)^4$

$= 16x^4 + 96x^3y + 216x^2y^2 + 216xy^3 + 81y^4$

→ The coefficient ${}^nC_r$ occuring in the binomial theorem are known as binomial coefficients.

→ There are $(n+1)$ terms in the expansion of $(a+b)^n$.

Eg: Compute $(98)^5$ using binomial theorem

ans:) $(98)^5 = (100-2)^5$

$= {}^5C_0(100)^5 - {}^5C_1(100)^4(2) + {}^5C_2(100)^3(2)^2 - {}^5C_3(100)^2(2)^3$
$+ {}^5C_4(100)(2)^4 - {}^5C_5(2)^5$

$= 399207968$ $\qquad$ $= 9039207968$

→ General term of binomial expansion

$(r+1)^{th}$ term,

$T_{r+1} = {}^nC_r\, a^{n-r} b^r \rightarrow (a+b)^n$

$T_{r+1} = (-1)^r\, {}^nC_r\, a^{n-r} b^r \rightarrow (a-b)^n$

Eg:- Find the 7^{th} term of the expansion of $(x - 3/x^2)^{20}$

ans) $r+1 = 7$, $n = 20$

$\therefore r = 6$

$T_7 = {}^{20}C_6\, x^{20-6} (3/x^2)^6 (-1)^6$

$= {}^{20}C_6\, x^{14} \dfrac{3^6}{x^{12}} = {}^{20}C_6\, 3^6 x^2$

Eg:- Find the coefficient of x^6y^3 in the expansion of $(x+2y)^9$

ans) $T_{r+1} = {}^9C_r\, x^{9-r} (2y)^r$ $\quad x^6 = x^{9-r}$

$9 - r = 6$

$r = 3$

$T_4 = {}^9C_3\, x^6 2^3 y^3$

$= {}^9C_3\, 2^3 x^6 y^3 = 672\, x^6 y^3$

→ Middle term(s) of $(a+b)^n$

→ If n is even, there is only one middle term

$= T_{\frac{n}{2}+1}$

Eg:- Middle term of $(x+2y)^8$ in the expansion

$= T_{\frac{8}{2}+1} = T_5$

→ If n is odd there are two middle terms

$= T_{\frac{n-1}{2}+1}\ ,\ T_{\frac{n+1}{2}+1}$

Eg:- Middle term in the expansion of $(2x-y)^7$ is

$$T_{\frac{7-1}{2}+1}, \quad T_{\frac{7+1}{2}+1}$$

$$= T_4, T_5$$

→ Term independent of x

Find the term independent of x in $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$

an) $T_{r+1} = {}^9C_r \left(\frac{3}{2}x^2\right)^{9-r} \left(\frac{1}{3x}\right)^r (-1)^r$

$$= (-1)^r\, {}^9C_r \left(\frac{3}{2}\right)^{9-r} (x^2)^{9-r} \left(\frac{1}{3}\right)^r (x^{-r}) \qquad x^{18-2r-r}$$

$$= (-1)^r\, {}^9C_r \left(\frac{3}{2}\right)^{9-r} \left(\frac{1}{3}\right)^r x^{18-2r-r}$$

$18-2r-r$

$18-3r=0$

$r = 18/3 = 6$

$$T_7 = {}^9C_6 \left(\frac{3}{2}\right)^3 \times \left(\frac{1}{3}\right)^6$$

$$= 7/18.$$

→ Multinomial theorem

For positive integers n, t, the coefficient of $x_1^{n_1} x_2^{n_2} x_3^{n_3} \cdots x_t^{n_t}$ in the expansion of $(x_1+x_2+x_3+\cdots+x_t)^n$ is

$$\frac{n!}{n_1!\,n_2!\,n_3!\cdots n_t!}$$

where each n_i is an integer with $0 \le n_i \le n$, for all $1 \le i \le t$,

and $n_1+n_2+n_3+\cdots+n_t = n$.

Tutorial

1) Determine the coefficient of x^9y^3 in the expansions of

(a) $(x+y)^{12}$ (b) $(x+2y)^{12}$ (c) $(2x-3y)^{12}$

ans:) $T_{r+1} = {}^nC_r a^{n-r} b^r$

a) $T_{r+1} = {}^{12}C_r x^{12-r} y^r$

$12-r=9$

$r=3$

$T_4 = {}^{12}C_3 x^9 y^3$

$= \underline{\underline{220}} x^9y^3$

(or) $\frac{12!}{9! \times 3!} = \underline{\underline{220}}$

b) ${}^{12}C_3 x^{12-r} (2y)^r$

${}^{12}C_3 \times 2^3 \times x^9 \times y^3$

$= \underline{\underline{1760}} x^9y^3$

c) $(-1)^3 \, {}^{12}C_3 (2x)^9 (3y)^3$

$= -{}^{12}C_3 2^9 3^3 x^9 y^3$

$= \underline{\underline{-3041280}}$

2) Determine the sum of all the coefficients in the expansion of (a) $(x+y)^3$ (b) $(x+y)^{10}$

ans:) (a) $(x+y)^3$ $n=3$ ${}^3C_0x^3 + {}^3C_1x^2y + {}^3C_2xy^2 + {}^3C_3y^3$

$= 2^3 = \underline{\underline{8}}$

$\left\{ {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots {}^nC_n = 2^n \right\}$

(b) $(x+y)^{10} = 2^{10} = 1024$